

Overfitting in Machine Learning

Overfitting refers to a model that models the training data too well.

Overfitting happens when a model learns the detail and noise in the training data to the extent that it negatively impacts the performance of the model on new data.

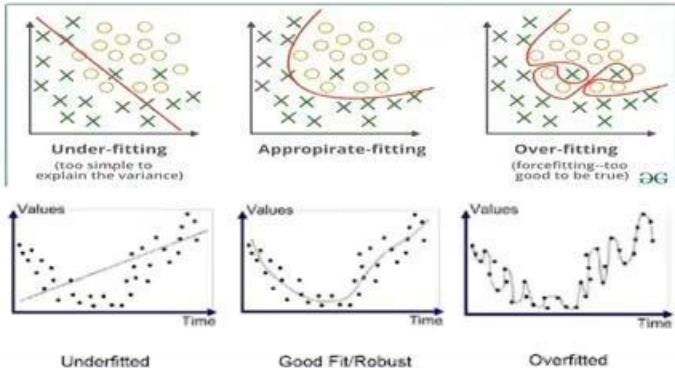
Underfitting in Machine Learning

Underfitting refers to a model that can neither model the training data nor generalize to new data.

An underfit machine learning model is not a suitable model and will be obvious as it will have poor performance on the training data.

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Examples



In brief

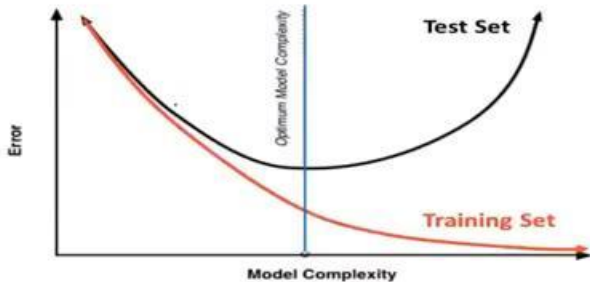
- **Overfitting:** Good performance on the training data, poor generalization to other data.
- **Underfitting:** Poor performance on the training data and poor generalization to other data

Regularization as solution for overfitting

1. What is Regularization?
2. How does Regularization help in reducing Overfitting?
3. Different Regularization techniques in Deep Learning
 1. Dropout
 2. Data augmentation

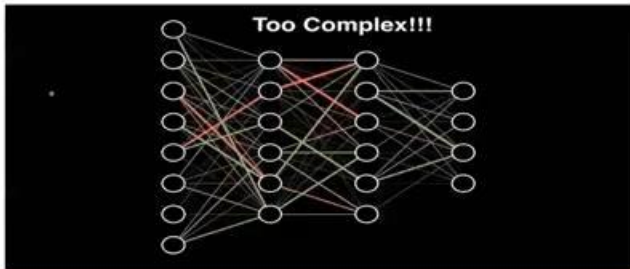
Problem of overfitting

Training Vs. Test Set Error



Complex network

If you've built a neural network before, you know how complex they are. This makes them more prone to overfitting.

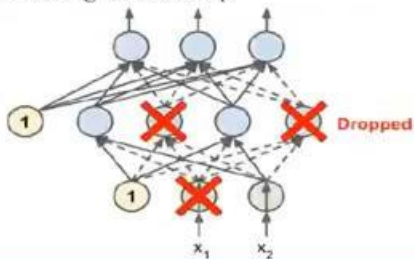


Regularization

- ❑ **Regularization** is a technique which makes slight modifications to the learning algorithm such that the model generalizes better.
- ❑ This in turn improves the model's performance on the unseen data as well.

Dropout

- It is a fairly simple algorithm: at every training step, every neuron (including the input neurons but excluding the output neurons) has a probability p of being temporarily “dropped out,” meaning it will be entirely ignored during this training step, but it may be active during the next step



Data Augmentation

- ❑ Consists of generating new training (rotating, resizing, flipping, and cropping) instances from existing ones, artificially boosting the size of the training set.
- ❑ This will reduce overfitting, making this a regularization technique. The trick is to generate realistic training instances.

